

Introduction to Homotopy Type Theory

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1 Dependent type theory

1.1 Element conversion

1.1.1 Element conversion rule

$$\frac{\frac{\Gamma \vdash A \doteq A' \text{ type}}{\Gamma \vdash A' \doteq A \text{ type}} \quad \frac{\Gamma \vdash A \text{ type}}{\Gamma, x : A' \vdash x : A'}}{\frac{\Gamma, x : A \vdash x : A' \quad \Gamma \vdash a : A}{\Gamma \vdash a : A'}}$$

1.1.2 Congruence rule for element conversion

$$\frac{\frac{\Gamma \vdash A \doteq A' \text{ type}}{\Gamma \vdash A' \doteq A \text{ type}} \quad \frac{\Gamma \vdash A \text{ type}}{\Gamma, x : A' \vdash x : A'}}{\frac{\Gamma, x : A \vdash x : A' \quad \Gamma \vdash a \doteq b : A}{\Gamma \vdash a \doteq b : A'}}$$

2 Dependent function types

2.1 Extensionality principle

$$\frac{\frac{\Gamma \vdash f : \prod_{(x:A)} B(x)}{\Gamma \vdash \lambda x. f(x) \doteq f : \prod_{(x:A)} B(x)} \quad \vdots}{\frac{\Gamma \vdash f \doteq \lambda x. f(x) : \prod_{(x:A)} B(x) \quad \Gamma \vdash \lambda x. f(x) \doteq g : \prod_{(x:A)} B(x)}{\Gamma \vdash f \doteq g : \prod_{(x:A)} B(x)}}$$
$$\frac{\frac{\Gamma, x : A \vdash f(x) \doteq g}{\Gamma \vdash \lambda x. f(x) \doteq \lambda x. g} \quad \frac{\Gamma \vdash g : \prod_{(x:A)} B(x)}{\Gamma \vdash \lambda x. g(x) \doteq g : \prod_{(x:A)} B(x)}}{\Gamma \vdash \lambda x. f(x) \doteq g : \prod_{(x:A)} B(x)}$$

2.2 Right unit law

$$\frac{\frac{\Gamma \vdash A \text{ type}}{\Gamma, x : A \vdash id_A(x) \doteq x : A} \quad \frac{\Gamma \vdash f : A \rightarrow B}{\Gamma, x' : A \vdash f(x') : B}}{\frac{\Gamma, x : A \vdash f(id_A(x)) \doteq f(x) : B}{\Gamma \vdash \lambda x. f(id_A(x)) \doteq \lambda x. f(x) : B}}$$
$$\frac{\vdots \quad \frac{\Gamma \vdash f : A \rightarrow B}{\Gamma \vdash \lambda x. f(x) \doteq f : A \rightarrow B}}{\frac{\Gamma \vdash \lambda x. f(id_A(x)) \doteq \lambda x. f(x) : B \quad \Gamma \vdash \lambda x. f(x) \doteq f : A \rightarrow B}{\Gamma \vdash \lambda x. f(id_A(x)) \doteq f : A \rightarrow B}}$$

2.3 Constant map

2.3.1 Construction

$$\frac{\frac{\frac{\Gamma \vdash B \text{ type}}{\Gamma, y : B \vdash y : B} \quad \Gamma \vdash A \text{ type}}{\Gamma, y : B, x : A \vdash y : B}}{\Gamma, y : B \vdash \lambda x. y : A \rightarrow B}}{\Gamma, y : B \vdash \text{const}_y := \lambda x. y : A \rightarrow B}$$

2.3.2 Right composition

$$\frac{\frac{\frac{\Gamma \vdash B \text{ type}}{\Gamma, z : C \vdash \text{const}_z \doteq \lambda y. z : B \rightarrow C}}{\Gamma, z : C, y : B \vdash \text{const}_z(y) \doteq (\lambda y. z)(y)} \quad \frac{\Gamma \vdash C \text{ type}}{\Gamma, y : B, z : C \vdash (\lambda y. z)(y) \doteq z : B}}{\Gamma, z : C, y : B \vdash \text{const}_z(y) \doteq z : C}$$

$$\frac{\frac{\vdots}{\Gamma, z : C, y : B \vdash \text{const}_z(y) \doteq z : C} \quad \frac{\Gamma \vdash f : A \rightarrow B}{\Gamma, x : A \vdash f(x) : B}}{\Gamma, z : C, x : A \vdash \text{const}_z(f(x)) \doteq z : C}$$

$$\frac{\frac{\vdots}{\Gamma, z : C, x : A \vdash \text{const}_z(f(x)) \doteq z : C} \quad \frac{\vdots}{\Gamma, z : C, x : A \vdash \text{const}_z(x) \doteq z : C}}{\Gamma, z : C, x : A \vdash \text{const}_z(f(x)) \doteq \text{const}_z(x) : C}$$

$$\frac{\Gamma, z : C \vdash \lambda x. \text{const}_z(f(x)) \doteq \lambda x. \text{const}_z(x) : A \rightarrow C}{\Gamma, z : C \vdash \text{const}_z \circ f \doteq \text{const}_z : A \rightarrow C}$$

2.3.3 Left composition

$$\frac{\frac{\Gamma \vdash A \text{ type} \quad \Gamma \vdash B \text{ type}}{\Gamma, y : B, x : A \vdash \text{const}_y(x) \doteq y : B} \quad \frac{\Gamma \vdash g : B \rightarrow C}{\Gamma, y' : B \vdash f(y') : C}}{\Gamma, y : B, x : A \vdash g(\text{const}_y(x)) \doteq g(y) : C}$$

$$\frac{\frac{\Gamma \vdash A \text{ type} \quad \Gamma \vdash C \text{ type}}{\Gamma, z : C, x : A \vdash \text{const}_z(x) \doteq z : C} \quad \frac{\Gamma \vdash g : B \rightarrow C}{\Gamma, y : B \vdash g(y) : C}}{\Gamma, y : B, x : A \vdash \text{const}_{g(y)}(x) \doteq g(y) : C}$$

$$\frac{\frac{\vdots}{\Gamma, y : B, x : A \vdash g(\text{const}_y(x)) \doteq g(y) : C} \quad \frac{\vdots}{\Gamma, y : B, x : A \vdash \text{const}_{g(y)}(x) \doteq g(y) : C}}{\Gamma, y : B, x : A \vdash g(\text{const}_y(x)) \doteq \text{const}_{g(y)}(x) : C}$$

$$\frac{\Gamma, y : B, \vdash \lambda x. g(\text{const}_y(x)) \doteq \lambda x. \text{const}_{g(y)}(x) : C}{\Gamma, y : B \vdash g \circ \text{const}_y \doteq \text{const}_{g(y)} : A \rightarrow C}$$

2.4 Swap function

2.4.1 Construction

$$\begin{array}{c}
\frac{\Gamma \vdash A \text{ type} \quad \frac{\Gamma \vdash B \text{ type} \quad \Gamma, x : A, y : B \vdash C(x, y) \text{ type}}{\Gamma, x : A \vdash \prod_{(y:B)} C(x, y) \text{ type}}}{\Gamma \vdash \prod_{(x:A)} \prod_{(y:B)} C(x, y) \text{ type}} \\
\frac{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y) \vdash f : \prod_{(x:A)} \prod_{(y:B)} C(x, y)}{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y), x : A \vdash f(x) : \prod_{(y:B)} C(x, y)} \\
\frac{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y), x : A, y : B \vdash f(x, y) : C(x, y)}{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y), y : B, x : A \vdash f(x, y) : C(x, y)} \\
\frac{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y), y : B \vdash \lambda x. f(x, y) : \prod_{(x:A)} C(x, y)}{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y) \vdash \lambda y. \lambda x. f(x, y) : \prod_{(y:B)} \prod_{(x:A)} C(x, y)} \\
\frac{\Gamma \vdash \lambda f. \lambda y. \lambda x. f(x, y) : (\prod_{(x:A)} \prod_{(y:B)} C(x, y)) \rightarrow (\prod_{(y:B)} \prod_{(x:A)} C(x, y))}{\Gamma \vdash \sigma := \lambda f. \lambda y. \lambda x. f(x, y) : (\prod_{(x:A)} \prod_{(y:B)} C(x, y)) \rightarrow (\prod_{(y:B)} \prod_{(x:A)} C(x, y))}
\end{array}$$

2.4.2 Swap is idempotent

$$\begin{array}{c}
\frac{\Gamma \vdash A \text{ type} \quad \Gamma \vdash B \text{ type} \quad \Gamma, x : A, y : B \vdash C(x, y) \text{ type}}{\Gamma \vdash \sigma : (\prod_{(y:B)} \prod_{(x:A)} C(x, y)) \rightarrow (\prod_{(x:A)} \prod_{(y:B)} C(x, y))} \\
\frac{\Gamma, g : \prod_{(y:B)} \prod_{(x:A)} C(x, y) \vdash \sigma(g) : \prod_{(x:A)} \prod_{(y:B)} C(x, y)}{\vdots} \\
\frac{\Gamma \vdash A \text{ type} \quad \Gamma \vdash B \text{ type} \quad \Gamma, x : A, y : B \vdash C(x, y) \text{ type}}{\Gamma \vdash \sigma \doteq \lambda f. \lambda y. \lambda x. f(x, y) : (\prod_{(x:A)} \prod_{(y:B)} C(x, y)) \rightarrow (\prod_{(y:B)} \prod_{(x:A)} C(x, y))} \\
\frac{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y) \vdash \sigma(f) \doteq \lambda y. \lambda x. f(x, y) : \prod_{(y:B)} \prod_{(x:A)} C(x, y)}{\vdots} \\
\frac{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y) \vdash \sigma(\sigma(f)) \doteq \sigma(\lambda y. \lambda x. f(x, y)) : \prod_{(x:A)} \prod_{(y:B)} C(x, y)}{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y) \vdash \sigma(\sigma(f)) \doteq \lambda x. \lambda y. f(x, y) : \prod_{(x:A)} \prod_{(y:B)} C(x, y)} \\
\frac{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y), x : A, y : B \vdash \sigma(\sigma(f, x, y)) \doteq f(x, y) : C(x, y)}{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y), x : A, y : B \vdash \sigma(\sigma(f, x, y)) \doteq id(f)(x, y) : C(x, y)} \\
\frac{\Gamma, f : \prod_{(x:A)} \prod_{(y:B)} C(x, y) \vdash \sigma(\sigma(f)) \doteq id(f) : \prod_{(x:A)} \prod_{(y:B)} C(x, y)}{\Gamma \vdash \sigma \circ \sigma \doteq id : (\prod_{(x:A)} \prod_{(y:B)} C(x, y)) \rightarrow (\prod_{(x:A)} \prod_{(y:B)} C(x, y))}
\end{array}$$

3 The natural numbers

3.1 Multiplication and exponentiation

3.1.1 Multiplication

$$mul_{\mathbb{N}}(m, 0_{\mathbb{N}}) := 0_{\mathbb{N}} \quad (1)$$

$$mul_{\mathbb{N}}(m, succ_{\mathbb{N}}(n)) := m + mul_{\mathbb{N}}(m, n) \quad (2)$$

$$\begin{array}{c}
\frac{\vdash \mathbb{N} \text{ type}}{\vdash \mathbb{N} \text{ type} \quad \vdash 0_{\mathbb{N}} : \mathbb{N}} \\
\frac{m : \mathbb{N} \vdash 0_{\mathbb{N}} : \mathbb{N}}{m : \mathbb{N} \vdash mul\text{-}zero_{\mathbb{N}}(m) := 0_{\mathbb{N}} : \mathbb{N}} \\
\frac{\vdash add_{\mathbb{N}} : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}}{m : \mathbb{N} \vdash add_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N}} \\
\frac{m : \mathbb{N}, n : \mathbb{N} \vdash add_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N}}{m : \mathbb{N} \vdash \lambda n. add_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}} \\
\frac{m : \mathbb{N} \vdash \lambda n. add_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}}{m : \mathbb{N} \vdash mul\text{-}succ_{\mathbb{N}} := \lambda n. add_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}}
\end{array}$$

$$\begin{array}{c}
\vdots \\
\hline
m : \mathbb{N} \vdash \text{mul-zero}_{\mathbb{N}}(m) := 0_{\mathbb{N}} : \mathbb{N} \quad m : \mathbb{N} \vdash \text{mul-succ}_{\mathbb{N}} := \lambda n. \text{add}_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \text{ind}_{\mathbb{N}}(\text{mul-zero}_{\mathbb{N}}(m), \text{mul-succ}_{\mathbb{N}}(m)) : \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \text{mul}_{\mathbb{N}}(m) := \text{ind}_{\mathbb{N}}(\text{mul-zero}_{\mathbb{N}}(m), \text{mul-succ}_{\mathbb{N}}(m)) : \mathbb{N} \rightarrow \mathbb{N}
\end{array}$$

3.1.2 Exponentiation

$$\text{exp}_{\mathbb{N}}(m, 0_{\mathbb{N}}) := 1_{\mathbb{N}} \quad (3)$$

$$\text{exp}_{\mathbb{N}}(m, \text{succ}_{\mathbb{N}}(n)) := m \cdot \text{exp}_{\mathbb{N}}(m, n) \quad (4)$$

$$\begin{array}{c}
\vdots \\
\hline
\vdash \mathbb{N} \text{ type} \\
\hline
\vdash 0_{\mathbb{N}} : \mathbb{N} \\
\hline
\vdash 1_{\mathbb{N}} : \mathbb{N} \\
\hline
\vdash \mathbb{N} \text{ type} \quad \vdash 1_{\mathbb{N}} : \mathbb{N} \\
\hline
m : \mathbb{N} \vdash 1_{\mathbb{N}} : \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \text{exp-zero}_{\mathbb{N}}(m) := 1_{\mathbb{N}} : \mathbb{N} \\
\hline
\vdash \text{mul}_{\mathbb{N}} : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \text{mul}_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N}, n : \mathbb{N} \vdash \text{mul}_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \lambda n. \text{mul}_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \text{exp-succ}_{\mathbb{N}} := \lambda n. \text{mul}_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \\
\hline
\vdots \\
\hline
m : \mathbb{N} \vdash \text{exp-zero}_{\mathbb{N}}(m) := 0_{\mathbb{N}} : \mathbb{N} \quad m : \mathbb{N} \vdash \text{exp-succ}_{\mathbb{N}} := \lambda n. \text{mul}_{\mathbb{N}}(m) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \text{ind}_{\mathbb{N}}(\text{exp-zero}_{\mathbb{N}}(m), \text{exp-succ}_{\mathbb{N}}(m)) : \mathbb{N} \rightarrow \mathbb{N} \\
\hline
m : \mathbb{N} \vdash \text{exp}_{\mathbb{N}}(m) := \text{ind}_{\mathbb{N}}(\text{exp-zero}_{\mathbb{N}}(m), \text{exp-succ}_{\mathbb{N}}(m)) : \mathbb{N} \rightarrow \mathbb{N}
\end{array}$$

3.2 Min and max functions

3.2.1 Min

$$\text{min}_{\mathbb{N}}(0_{\mathbb{N}}, n) := 0_{\mathbb{N}} \quad (5)$$

$$\text{min}_{\mathbb{N}}(m, 0_{\mathbb{N}}) := 0_{\mathbb{N}} \quad (6)$$

$$\text{min}_{\mathbb{N}}(\text{succ}_{\mathbb{N}}(m), \text{succ}_{\mathbb{N}}(n)) := \text{succ}_{\mathbb{N}}(\text{min}_{\mathbb{N}}(m, n)) \quad (7)$$

$$\text{min}_{\mathbb{N}} \doteq \text{ind}_{\mathbb{N}}(\text{const}_{0_{\mathbb{N}}}, \lambda m. \lambda f. \lambda n. \text{ind}_{\mathbb{N}}(0_{\mathbb{N}}, \lambda x. \lambda y. \text{succ}_{\mathbb{N}}(f(y)), n))$$

3.2.2 Max

$$\text{max}_{\mathbb{N}}(0_{\mathbb{N}}, n) := n \quad (8)$$

$$\text{max}_{\mathbb{N}}(m, 0_{\mathbb{N}}) := m \quad (9)$$

$$\text{max}_{\mathbb{N}}(\text{succ}_{\mathbb{N}}(m), \text{succ}_{\mathbb{N}}(n)) := \text{succ}_{\mathbb{N}}(\text{max}_{\mathbb{N}}(m, n)) \quad (10)$$

$$\text{max}_{\mathbb{N}} \doteq \text{ind}_{\mathbb{N}}(\text{id}_{\mathbb{N}}, \lambda m. \lambda f. \lambda n. \text{ind}_{\mathbb{N}}(\text{succ}_{\mathbb{N}}(m), \lambda x. \lambda y. \text{succ}_{\mathbb{N}}(f(y)), n))$$

3.3 Iterator

3.3.1 Triangle numbers

$$\text{tri} := \text{ind}_{\mathbb{N}}(0_{\mathbb{N}}, \lambda x. \lambda y. \text{succ}_{\mathbb{N}}(x) + y)$$

3.3.2 Factorial

$$\text{fac} := \text{ind}_{\mathbb{N}}(0_{\mathbb{N}}, \lambda x. \lambda y. \text{succ}_{\mathbb{N}}(x) \cdot y)$$

3.4 Binomial coefficient

$$\binom{-}{-} := \text{ind}_{\mathbb{N}}(\text{ind}_{\mathbb{N}}(1_{\mathbb{N}}, \lambda x. \lambda y. 0_{\mathbb{N}}), \lambda n. \lambda f. \lambda k. \text{ind}_{\mathbb{N}}(1_{\mathbb{N}}, \lambda x. \lambda y. f(x) + f(\text{succ}_{\mathbb{N}}(x))))$$

3.5 Fibonacci sequence

$$F := \lambda n. \text{pr}_1(\text{ind}_{\mathbb{N}}((0_{\mathbb{N}}, 1_{\mathbb{N}}), \lambda n. \lambda p. (\text{pr}_2(p), \text{pr}_1(p) + \text{pr}_2(p)), n)) \text{ where :}$$

$$(-, -) := \lambda a. \lambda b. \lambda f. f(a, b) : \mathbb{N} \rightarrow \mathbb{N} \rightarrow (\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}) \rightarrow \mathbb{N} \quad (11)$$

$$\text{pr}_1 := \lambda p. p(\lambda a. \lambda b. a) : ((\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}) \rightarrow \mathbb{N}) \rightarrow \mathbb{N} \quad (12)$$

$$\text{pr}_2 := \lambda p. p(\lambda a. \lambda b. b) : ((\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}) \rightarrow \mathbb{N}) \rightarrow \mathbb{N} \quad (13)$$

3.6 Division by two rounded down

$$\text{div}_2(0_{\mathbb{N}}) := 0_{\mathbb{N}} \quad (14)$$

$$\text{div}_2(1_{\mathbb{N}}) := 0_{\mathbb{N}} \quad (15)$$

$$\text{div}_2(\text{succ}_{\mathbb{N}}(\text{succ}_{\mathbb{N}}(n))) := \text{succ}_{\mathbb{N}}(\text{div}_2(n)) \quad (16)$$

$$\text{div}_2 := \text{ind}_{\mathbb{N}}(0_{\mathbb{N}}, \text{ind}_{\mathbb{N}}(0_{\mathbb{N}}, \lambda x. \lambda y. \text{succ}_{\mathbb{N}}(y)))$$

4 More inductive types

4.1 Operations on the Integers

4.1.1 Predecessor

$$\text{pred}_{\mathbb{Z}}(-1_{\mathbb{Z}}) := \text{in-neg}(1_{\mathbb{N}}) \quad (17)$$

$$\text{pred}_{\mathbb{Z}}(\text{in-neg}(\text{succ}_{\mathbb{N}}(n))) := \text{in-neg}(\text{succ}_{\mathbb{N}}(\text{succ}_{\mathbb{N}}(n))) \quad (18)$$

$$\text{pred}_{\mathbb{Z}}(0_{\mathbb{Z}}) := -1_{\mathbb{Z}} \quad (19)$$

$$\text{pred}_{\mathbb{Z}}(1_{\mathbb{Z}}) := 0_{\mathbb{Z}} \quad (20)$$

$$\text{pred}_{\mathbb{Z}}(\text{in-pos}(\text{succ}_{\mathbb{N}}(n))) := \text{in-pos}(n) \quad (21)$$

4.1.2 Addition and negation

$$\text{add}_{\mathbb{Z}}(-1_{\mathbb{Z}}, n) := \text{pred}_{\mathbb{Z}}(n) \quad (22)$$

$$\text{add}_{\mathbb{Z}}(\text{in-neg}(\text{succ}_{\mathbb{N}}(m)), n) := \text{pred}_{\mathbb{Z}}(\text{add}_{\mathbb{Z}}(\text{in-neg}(m), n)) \quad (23)$$

$$\text{add}_{\mathbb{Z}}(0_{\mathbb{Z}}, n) := n \quad (24)$$

$$\text{add}_{\mathbb{Z}}(1_{\mathbb{Z}}, n) := \text{succ}_{\mathbb{Z}}(n) \quad (25)$$

$$\text{add}_{\mathbb{Z}}(\text{in-pos}(\text{succ}_{\mathbb{N}}(m)), n) := \text{succ}_{\mathbb{Z}}(\text{add}(\text{in-pos}(m), n)) \quad (26)$$

$$\text{neg}_{\mathbb{Z}}(\text{in-neg}(n)) := \text{in-pos}(n) \quad (27)$$

$$\text{neg}_{\mathbb{Z}}(0_{\mathbb{Z}}) := 0_{\mathbb{Z}} \quad (28)$$

$$\text{neg}_{\mathbb{Z}}(\text{in-pos}(n)) := \text{in-neg}(n) \quad (29)$$

4.1.3 Multiplication

$$\text{mul}_{\mathbb{Z}}(\text{in-neg}(m), n) := \text{neg}_{\mathbb{Z}}(\text{mul}(\text{in-pos}(m), n)) \quad (30)$$

$$\text{mul}_{\mathbb{Z}}(0_{\mathbb{Z}}, n) := 0_{\mathbb{Z}} \quad (31)$$

$$\text{mul}_{\mathbb{Z}}(1_{\mathbb{Z}}, n) := n \quad (32)$$

$$\text{mul}_{\mathbb{Z}}(\text{in-pos}(\text{succ}_{\mathbb{N}}(m)), n) := \text{mul}_{\mathbb{Z}}(\text{in-pos}(m), n) + n \quad (33)$$

4.2 Boolean

4.2.1 Negation

$$\frac{\frac{\frac{\vdash \text{bool type}}{\vdash \text{true} : \text{bool}} \quad \frac{\vdash \text{bool type}}{\vdash \text{false} : \text{bool}}}{\vdash \text{ind-bool}(\text{true}, \text{false}) : \text{bool} \rightarrow \text{bool}}}{\vdash \text{neg-bool} := \text{ind-bool}(\text{true}, \text{false}) : \text{bool} \rightarrow \text{bool}}$$

4.2.2 Conjunction

$$\frac{\frac{\vdash \text{bool type} \quad \frac{\vdash \text{bool type}}{\vdash \text{false} : \text{bool}}}{\vdash \text{const}_{\text{false}} : \text{bool} \rightarrow \text{bool}} \quad \frac{\vdash \text{bool type}}{\vdash \text{id}_{\text{bool}} : \text{bool} \rightarrow \text{bool}}}{\frac{\vdash \text{ind-bool}(\text{const}_{\text{false}}, \text{id}_{\text{bool}})}{\vdash - \wedge - := \text{ind-bool}(\text{const}_{\text{false}}, \text{id}_{\text{bool}})}}$$

4.2.3 Disjunction

$$\frac{\frac{\vdash \text{bool type}}{\vdash \text{id}_{\text{bool}} : \text{bool} \rightarrow \text{bool}} \quad \frac{\vdash \text{bool type} \quad \frac{\vdash \text{bool type}}{\vdash \text{true} : \text{bool}}}{\vdash \text{const}_{\text{true}} : \text{bool} \rightarrow \text{bool}}}{\frac{\vdash \text{ind-bool}(\text{id}_{\text{bool}}, \text{const}_{\text{true}})}{\vdash - \vee - := \text{ind-bool}(\text{id}_{\text{bool}}, \text{const}_{\text{true}})}}$$

4.3 Type theoretic proofs of the constructive tautologies

4.3.1 Basic

$$\frac{\frac{\frac{\vdots}{\Gamma \vdash P \times \neg P \text{ type}}}{\Gamma, t : P \times \neg P \vdash t : P \times \neg P} \quad \frac{\Gamma, t : P \times \neg P \vdash \text{pr}_2(t) : \neg P}{\Gamma, t : P \times \neg P, p : P \vdash \text{pr}_2(t)(p) : \emptyset}}{\frac{\frac{\vdots}{\Gamma, t : P \times \neg P \vdash \text{pr}_1(t) : P}}{\Gamma, t : P \times \neg P \vdash \text{pr}_2(t)(\text{pr}_1(t)) : \emptyset}}}{\Gamma \vdash \lambda t. \text{pr}_2(t)(\text{pr}_1(t)) : \neg(P \times \neg P)}$$

$$\frac{\frac{\frac{\vdots}{\Gamma \vdash P \leftrightarrow \neg P \text{ type}}}{\Gamma, t : P \leftrightarrow \neg P \vdash t : P \leftrightarrow \neg P} \quad \frac{\Gamma, t : P \leftrightarrow \neg P \vdash \text{pr}_1(t) : P \rightarrow \neg P}{\Gamma, t : P \leftrightarrow \neg P, p : P \vdash \text{pr}_1(t)(p, p) : \emptyset}}{\frac{\Gamma, t : P \leftrightarrow \neg P \vdash \lambda p. \text{pr}_1(t)(p, p) : \neg P}{\Gamma, t : P \leftrightarrow \neg P \vdash \hat{p} := \lambda p. \text{pr}_1(t)(p, p) : \neg P}}$$

$$\frac{\frac{\frac{\vdots}{\Gamma, t : P \leftrightarrow \neg P \vdash \text{pr}_2(t) : \neg P \rightarrow P}}{\Gamma, t : P \leftrightarrow \neg P, \tilde{p} : \neg P \vdash \text{pr}_2(t)(\tilde{p}) : P} \quad \frac{\vdots}{\Gamma, t : P \leftrightarrow \neg P \vdash \hat{p} : \neg P}}{\frac{\Gamma, t : P \leftrightarrow \neg P \vdash \text{pr}_2(t)(\hat{p}) : P}{\Gamma, t : P \leftrightarrow \neg P \vdash \hat{p}(\text{pr}_2(t)(\hat{p})) : \emptyset}}}{\Gamma \vdash \lambda t. \hat{p}(\text{pr}_2(t)(\hat{p})) : \neg(P \leftrightarrow \neg P)}$$

4.3.2 Double negation monad

$$\begin{array}{c}
\frac{\frac{\Gamma \vdash P \text{ type}}{\Gamma, p : P \vdash p : P} \quad \frac{\Gamma \vdash \neg P \text{ type}}{\Gamma, \tilde{p} : \neg P \vdash \tilde{p} : \neg P}}{\frac{\Gamma, p : P, \tilde{p} : \neg P \vdash \tilde{p}(p) : \emptyset}}{\Gamma \vdash \lambda p. \lambda \tilde{p}. \tilde{p}(p) : P \rightarrow \neg \neg P} \\
\\
\frac{\frac{\Gamma \vdash P \rightarrow Q \text{ type}}{\Gamma, f : P \rightarrow Q \vdash f : P \rightarrow Q} \quad \frac{\Gamma \vdash \neg Q \text{ type}}{\Gamma, \tilde{q} : \neg Q \vdash \tilde{q} : \neg Q} \quad \frac{\Gamma \vdash \neg \neg P \text{ type}}{\Gamma, p : \neg \neg P \vdash p : \neg \neg P}}{\frac{\Gamma, f : P \rightarrow Q, \tilde{q} : \neg Q \vdash \tilde{q} \circ f : \neg P}{\Gamma, f : P \rightarrow Q, p : \neg \neg P, \tilde{q} : \neg Q \vdash p(\tilde{q} \circ f) : \emptyset}}{\Gamma \vdash \lambda f. \lambda p. \lambda \tilde{q}. p(\tilde{q} \circ f) : (P \rightarrow Q) \rightarrow (\neg \neg P \rightarrow \neg \neg Q)} \\
\\
\frac{\frac{\Gamma \vdash P \rightarrow \neg \neg Q \text{ type}}{\Gamma, f : P \rightarrow \neg \neg Q \vdash f : P \rightarrow \neg \neg Q} \quad \frac{\Gamma \vdash \neg Q \text{ type}}{\Gamma, \tilde{q} : \neg Q \vdash \tilde{q} : \neg Q} \quad \frac{\Gamma \vdash \neg \neg P \text{ type}}{\Gamma, p : \neg \neg P \vdash p : \neg \neg P}}{\frac{\Gamma, f : P \rightarrow \neg \neg Q, \tilde{q} : \neg Q \vdash \text{swap}(f)(\tilde{q}) : \neg P}{\Gamma, f : P \rightarrow \neg \neg Q, p : \neg \neg P, \tilde{q} : \neg Q \vdash p(\text{swap}(f)(\tilde{q})) : \emptyset}}{\Gamma \vdash p(\text{swap}(f)(\tilde{q})) : (P \rightarrow \neg \neg Q) \rightarrow (\neg \neg P \rightarrow \neg \neg Q)}
\end{array}$$

4.3.3 Double negation of classical laws

$$\begin{array}{c}
\frac{\frac{\Gamma \vdash \neg(\neg \neg P \rightarrow P) \text{ type}}{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash u : \neg(\neg \neg P \rightarrow P)} \quad \frac{\frac{\Gamma \vdash P \text{ type}}{\Gamma, p : P \vdash p : P} \quad \Gamma \vdash \neg \neg P \text{ type}}{\Gamma, p : P \vdash \text{const}_p : \neg \neg P \rightarrow P}}{\frac{\Gamma, u : \neg(\neg \neg P \rightarrow P), p : P \vdash u(\text{const}_p) : \emptyset}}{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash \lambda p. u(\text{const}_p) : \neg P} \\
\\
\frac{\vdots}{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash \lambda p. u(\text{const}_p) : \neg P} \quad \frac{\Gamma \vdash \neg \neg P \text{ type}}{\Gamma, \hat{p} : \neg \neg P \vdash \hat{p} : \neg \neg P} \\
\frac{\Gamma, u : \neg(\neg \neg P \rightarrow P), \hat{p} : \neg \neg P \vdash \hat{p}(\lambda p. u(\text{const}_p)) : \emptyset}{\frac{\Gamma, u : \neg(\neg \neg P \rightarrow P), \hat{p} : \neg \neg P \vdash \text{ind}_\emptyset(\hat{p}(\lambda p. u(\text{const}_p))) : P}{\frac{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash \lambda \hat{p}. \text{ind}_\emptyset(\hat{p}(\lambda p. u(\text{const}_p))) : \neg \neg P \rightarrow P}{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash k := \lambda \hat{p}. \text{ind}_\emptyset(\hat{p}(\lambda p. u(\text{const}_p))) : \neg \neg P \rightarrow P}}} \\
\\
\frac{\Gamma \vdash \neg(\neg \neg P \rightarrow P) \text{ type}}{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash u : \neg(\neg \neg P \rightarrow P)} \quad \frac{\vdots}{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash k : \neg \neg P \rightarrow P} \\
\frac{\Gamma, u : \neg(\neg \neg P \rightarrow P) \vdash u(k) : \emptyset}{\Gamma \vdash \lambda u. u(k) : \neg \neg(\neg \neg P \rightarrow P)} \\
\\
\frac{\frac{\Gamma \vdash P \text{ type}}{\Gamma, p : P \vdash p : P} \quad \Gamma \vdash (P \rightarrow Q) \rightarrow P \text{ type}}{\Gamma, p : P \vdash \text{const}_p : ((P \rightarrow Q) \rightarrow P) \rightarrow P} \quad \frac{\Gamma \vdash \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P) \text{ type}}{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P) \vdash u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P)} \\
\frac{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P), p : P \vdash u(\text{const}_p) : \emptyset}{\frac{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P), p : P \vdash \text{ind}_\emptyset(u(\text{const}_p)) : Q}{\frac{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P) \vdash \lambda p. \text{ind}_\emptyset(u(\text{const}_p)) : P \rightarrow Q}{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P) \vdash k := \lambda p. \text{ind}_\emptyset(u(\text{const}_p)) : P \rightarrow Q}}} \\
\\
\frac{\Gamma \vdash (P \rightarrow Q) \rightarrow P \text{ type}}{\Gamma, f : (P \rightarrow Q) \rightarrow P \vdash f : (P \rightarrow Q) \rightarrow P} \quad \frac{\vdots}{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P) \vdash k : P \rightarrow Q} \\
\frac{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P), f : (P \rightarrow Q) \rightarrow P \vdash f(k) : P}{\frac{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P) \vdash \lambda f. f(k) : ((P \rightarrow Q) \rightarrow P) \rightarrow P}{\frac{\Gamma, u : \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P) \vdash u(\lambda f. f(k)) : \emptyset}{\Gamma \vdash \lambda u. u(\lambda f. f(k)) : \neg \neg(((P \rightarrow Q) \rightarrow P) \rightarrow P)}}}
\end{array}$$

Assume $u : \neg((P \rightarrow Q) + (Q \rightarrow P))$:
 Assume $q : Q$. There is $const_q : P \rightarrow Q$. Then $inl(const_q) : (P \rightarrow Q) + (Q \rightarrow P)$. And $u(inl(const_q)) : \emptyset$.
 So, $\lambda q.ind_{\emptyset}(u(inl(const_q))) : Q \rightarrow P$.
 Then $inr(\lambda q.ind_{\emptyset}(u(inl(const_q)))) : (P \rightarrow Q) + (Q \rightarrow P)$. And $u(inr(\lambda q.ind_{\emptyset}(u(inl(const_q)))))) : \emptyset$.
 Finally, $\lambda u.u(inr(\lambda q.ind_{\emptyset}(u(inl(const_q)))))) : \neg((P \rightarrow Q) + (Q \rightarrow P))$.

Assume $u : \neg(P + \neg P)$:
 Assume $p : P$. Then, we have $inl(p) : P + \neg P$. So $\lambda p.u(inl(p)) : \neg P$.
 Let $inr(\lambda p.u(inl(p))) : P + \neg P$. Finally, $\lambda u.u(inr(\lambda p.u(inl(p)))) : \neg(P + \neg P)$.

4.3.4 Show that

Assume $em : P + \neg P$ and $t : \neg\neg P$:
 Then $ind_+(id_P, \lambda \tilde{p}.ind_{\emptyset}(t(\tilde{p})), em) : P$.
 Finally, $dne := \lambda em.\lambda t.ind_+(id_P, \lambda \tilde{p}.ind_{\emptyset}(t(\tilde{p})), em) : (P + \neg P) \rightarrow (\neg\neg P \rightarrow P)$.

Split into two cases:

Case 1, $\neg\neg(Q \rightarrow P) \rightarrow (P + \neg P) \rightarrow (Q \rightarrow P)$:
 Assume $u : \neg\neg(Q \rightarrow P)$, $em : (P + \neg P)$ and $q : Q$:
 In em , if $p : P$ then $\lambda q.p : Q \rightarrow P$, else $\tilde{p} : \neg P$ then:
 Assume $f : Q \rightarrow P$ then $\lambda f.\tilde{p}(f(q)) : \neg(P \rightarrow Q)$.
 So, $ind_{\emptyset}(u(\lambda f.\tilde{p}(f(q)))) : P$ then $\lambda q.ind_{\emptyset}(u(\lambda f.\tilde{p}(f(q)))) : Q \rightarrow P$
 Finally, $\lambda u.\lambda em.ind_+(\lambda p.const_p, \lambda \tilde{p}.\lambda q.ind_{\emptyset}(u(\lambda f.\tilde{p}(f(q))))), em) : \neg\neg(Q \rightarrow P) \rightarrow (P + \neg P) \rightarrow (Q \rightarrow P)$
 Case 2, $((P + \neg P) \rightarrow (Q \rightarrow P)) \rightarrow \neg\neg(Q \rightarrow P)$:

$$\lambda f.\lambda u.u(f(inr(\lambda p.u(f(inl(p)))))) : ((P + \neg P) \rightarrow (Q \rightarrow P)) \rightarrow \neg\neg(Q \rightarrow P)$$

4.3.5 Double negation stable

Assume $t : \neg\neg\neg P$ and $p : P$:
 We have $\lambda u.u(p) : \neg\neg P$. Then $t(\lambda u.u(p)) : \emptyset$.
 Finally, $\lambda t.\lambda p.t(\lambda u.u(p)) : \neg\neg\neg P \rightarrow \neg P$.

Assume $t : \neg\neg(P \rightarrow \neg\neg Q)$, $p : P$, $u : \neg Q$, and $v : P \rightarrow \neg\neg Q$:
 Then $\lambda v.v(p, u) : \neg(P \rightarrow \neg\neg Q)$.
 Finally, $\lambda t.\lambda p.t(\lambda v.v(p, u)) : \neg\neg(P \rightarrow \neg\neg Q) \rightarrow (P \rightarrow \neg\neg Q)$

Assume $t : \neg\neg((\neg\neg P) \times (\neg\neg Q))$, $u : \neg P$:
 Assume $v : (\neg\neg P) \times (\neg\neg Q)$:
 Then $\lambda v.pr_1(v)(u) : \neg((\neg\neg P) \times (\neg\neg Q))$.
 Analogously, $\lambda w.t(\lambda v.pr_2(v)(w)) : \neg\neg Q$.
 Finally, $\lambda t.(\lambda u.t(\lambda v.pr_1(v)(u)), \lambda w.t(\lambda v.pr_2(v)(w))) : \neg\neg((\neg\neg P) \times (\neg\neg Q)) \rightarrow ((\neg\neg P) \times (\neg\neg Q))$

4.3.6 Show that

Split into 2 cases:
 Case 1: $\neg\neg(P \times Q) \rightarrow (\neg\neg P) \times (\neg\neg Q)$.
 Assume $t : \neg\neg(P \times Q)$, $u : \neg P$ and $v : P \times Q$.
 Then $\lambda v.u(pr_1(v)) : \neg(P \times Q)$.
 So, $\lambda u.t(\lambda v.u(pr_1(v))) : \neg\neg P$.
 Analogously, $\lambda w.t(\lambda v.w(pr_2(v))) : \neg\neg Q$.
 Finally, $\lambda t.(\lambda u.t(\lambda v.u(pr_1(v))), \lambda w.t(\lambda v.w(pr_2(v)))) : \neg\neg(P \times Q) \rightarrow (\neg\neg P) \times (\neg\neg Q)$.
 Case 2: $(\neg\neg P) \times (\neg\neg Q) \rightarrow \neg\neg(P \times Q)$.
 Assume $t : (\neg\neg P) \times (\neg\neg Q)$ and $u : \neg(P \times Q)$.
 Assume $p : P$ and $q : Q$, then $u(p, q) : \emptyset$, so $\lambda q.u(p, q) : \neg Q$.
 Then $\lambda p.pr_2(t)(\lambda q.u(p, q)) : \neg P$.
 Then $pr_1(\lambda p.pr_2(t)(\lambda q.u(p, q))) : \emptyset$.
 Finally, $\lambda t.\lambda u.pr_1(\lambda p.pr_2(t)(\lambda q.u(p, q))) : (\neg\neg P) \times (\neg\neg Q) \rightarrow \neg\neg(P \times Q)$.

Split into 2 cases:

Case 1: $\neg\neg(P + Q) \rightarrow \neg(\neg P \times \neg Q)$.

Assume $t : \neg\neg(P + Q)$ and $u : \neg P \times \neg Q$.
 Assume $v : P + Q$ then $\lambda v.ind_+(pr_1(u), pr_2(u), v) : \neg(P + Q)$.
 Finally, $\lambda t.\lambda u.t(\lambda v.ind_+(pr_1(u), pr_2(u), v)) : \neg\neg(P + Q) \rightarrow \neg(\neg P \times \neg Q)$.
 Case 2: $\neg(\neg P \times \neg Q) \rightarrow \neg\neg(P + Q)$.
 Assume $t : \neg(\neg P \times \neg Q)$ and $u : \neg(P + Q)$.
 Assume $p : P$ then $\lambda p.u(inl(p)) : \neg P$.
 Assume $q : Q$ then $\lambda q.u(inr(q)) : \neg Q$.
 Finally $\lambda t.\lambda u.t(\lambda p.u(inl(p)), \lambda q.u(inr(q))) : \neg(\neg P \times \neg Q) \rightarrow \neg\neg(P + Q)$.

Split into 2 cases:

Case 1: $\neg\neg(P \rightarrow Q) \rightarrow (\neg\neg P \rightarrow \neg\neg Q)$.
 Assume $t : \neg\neg(P \rightarrow Q)$, $u : \neg\neg P$ and $v : \neg Q$.
 Assume $w : P \rightarrow Q$ and $p : P$ then $\lambda p.v(w(p)) : \neg P$.
 Then $\lambda w.u(\lambda p.v(w(p))) : \neg(P \rightarrow Q)$.
 Finally, $\lambda t.\lambda u.\lambda v.t(\lambda w.u(\lambda p.v(w(p)))) : \neg\neg(P \rightarrow Q) \rightarrow (\neg\neg P \rightarrow \neg\neg Q)$.
 Case 2: $(\neg\neg P \rightarrow \neg\neg Q) \rightarrow \neg\neg(P \rightarrow Q)$.
 Assume $t : \neg\neg P \rightarrow \neg\neg Q$ and $u : \neg(P \rightarrow Q)$.
 Assume $v : \neg P$. We have $ind_\emptyset : \emptyset \rightarrow Q$ then $ind_\emptyset \circ v : P \rightarrow Q$.
 Then $\lambda v.u(ind_\emptyset \circ v) : \neg\neg P$.
 Assume $q : Q$ then $\lambda q.u(const_q) : \neg Q$.
 Finally, $\lambda t.\lambda u.t(\lambda v.u(ind_\emptyset \circ v), \lambda q.u(const_q)) : (\neg\neg P \rightarrow \neg\neg Q) \rightarrow \neg\neg(P \rightarrow Q)$.

4.4 List types

4.4.1 Induction principle and the computation rules

$$\begin{array}{c}
 \frac{\Gamma, l : list(A) \vdash P(l) \text{ type} \quad \Gamma \vdash p_{nil} : P(nil) \quad \Gamma \vdash p_{cons} : \prod_{(a:A)} \prod_{l:list(A)} P(l) \rightarrow P(cons(a, l))}{\Gamma \vdash ind-list(p_{nil}, p_{cons}) : \prod_{(l:list(A))} P(l)} \\
 \\
 \frac{\Gamma, l : list(A) \vdash P(l) \text{ type} \quad \Gamma \vdash p_{nil} : P(nil) \quad \Gamma \vdash p_{cons} : \prod_{(a:A)} \prod_{l:list(A)} P(l) \rightarrow P(cons(a, l))}{\Gamma \vdash ind-list(p_{nil}, p_{cons}, nil) \doteq p_{nil} : P(l)} \\
 \\
 \frac{\Gamma, l : list(A) \vdash P(l) \text{ type} \quad \Gamma \vdash p_{nil} : P(nil) \quad \Gamma \vdash p_{cons} : \prod_{(a:A)} \prod_{l:list(A)} P(l) \rightarrow P(cons(a, l))}{\Gamma \vdash ind-list(p_{nil}, p_{cons}, cons(a, l)) \doteq p_{cons}(a, l, ind-list(p_{nil}, p_{cons}, l)) : P(l)}
 \end{array}$$

4.4.2 Folding

$$fold-list(\mu, nil) := b \quad (34)$$

$$fold-list(\mu, cons(a, l)) := \mu(a, fold-list(\mu, l)) \quad (35)$$

$$fold-list := \lambda \mu.ind-list(b, \lambda a.\lambda l.\lambda c.\mu(a, c))$$

4.4.3 Mapping

$$map-list(\mu, nil) := nil \quad (36)$$

$$map-list(\mu, cons(a, l)) := cons(\mu(a), map-list(\mu, l)) \quad (37)$$

$$map-list := \lambda \mu.ind-list(nil, \lambda a.\lambda l.\lambda c.cons(\mu(a), c))$$

4.4.4 Length

$$length-list(nil) := 0_{\mathbb{N}} \quad (38)$$

$$length-list(cons(a, l)) := succ_{\mathbb{N}}(length-list(l)) \quad (39)$$

$$length-list := ind-list(0_{\mathbb{N}}, \lambda a.\lambda l.\lambda c.succ_{\mathbb{N}}(c))$$

4.4.5 Sum and product

$$sum-list := ind-list(0_{\mathbb{N}}, \lambda a. \lambda l. \lambda c. a + c)$$

$$product-list := ind-list(0_{\mathbb{N}}, \lambda a. \lambda l. \lambda c. a \cdot c)$$

4.4.6 Concatenation

$$concat-list(nil, l) := l \tag{40}$$

$$concat-list(cons(a, k), l) := cons(a, concat-list(k, l)) \tag{41}$$

$$concat-list := ind-list(id_{list(A)}, \lambda a. \lambda l. \lambda c. cons(a, c))$$

4.4.7 Flattening

$$flatten-list(nil) := nil \tag{42}$$

$$flatten-list(cons(a, l)) := concat-list(a, flatten-list(l)) \tag{43}$$

$$flatten-list := ind-list(nil, \lambda a. \lambda l. \lambda c. concat-list(a, c))$$

4.4.8 Reversion

$$reverse-list(nil) := nil \tag{44}$$

$$reverse-list(cons(a, l)) := concat-list(reverse-list(l), cons(a, nil)) \tag{45}$$

$$reverse-list := ind-list(nil, \lambda a. \lambda l. \lambda c. concat-list(c, cons(a, nil)))$$

5 Identity types